Exponential Functions

Up to now, our focus was that of *algebraic functions*. Algebraic functions have variable bases raised to constant exponents. We will now turn our attention to *exponential functions* and their inverses called *logarithmic functions*.

The form of an exponential function is

$$f(x) = b^x,$$

where b is an arbitrary positive real number not equal to 1, and x is any real number.

1) For each real number x, the quantity b^x is a unique real number.

2) The properties of rational exponents that you've studied before also hold for irrational exponents.

3) If $b^x = b^y$ and $b \neq 1$, then $x = y$

4) If $b > 1$, the function 'grows'

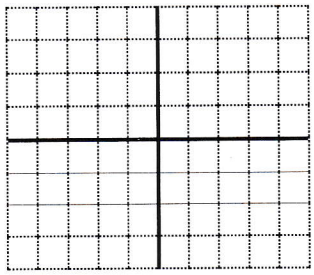
5) If $0 < b < 1$, the function 'decays'

<u>Example 1:</u>

Graph each set of functions on the same set of axes. Form ordered pairs of numbers using a table of values. Note that the base for each of these graphs is greater than one.

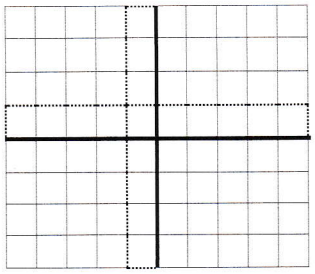
a) $y_1 = 3^x$, $y_2 = 2^x$

x	y



b) $y_1 = 3^{-x}$, $y_2 = 2^{-x}$

x	y



Properties of Graphs of Exponential Functions, $f(x) = b^x$

$b > 1$	$0 < b < 1$
Domain:	Domain:
Range:	Range:
Asymptote:	Asymptote:
y-intercept:	y-intercept:
x-intercept:	x-intercept:

Example 2:

Use your knowledge of translations and reflections to sketch each of the following. Specify the domain, range, intercept(s) and asymptote.

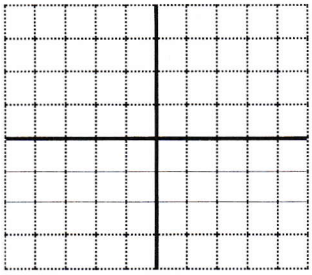
a) $y = 3^{x+1} + 1$

Domain:

Range:

Intercept(s):

Asymptote:



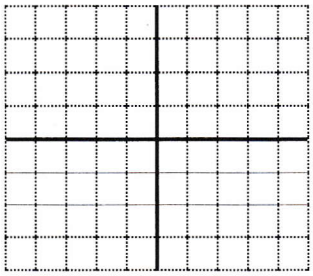
b) $y = 1 - 3^{1-x}$

Domain:

Range:

Intercept(s):

Asymptote:



From your observations of the graphs of the functions of the form $y = b^x$, do you think the exponential function is one-to-one? Why or why not?

The Exponential Function $y = e^x$

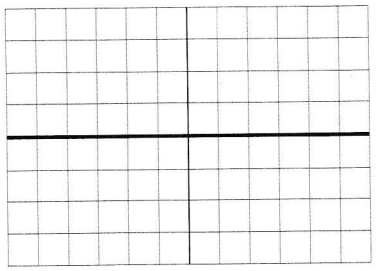
There is a very useful base named 'e' whose value is used in applications ranging from finance to physics. The value $e = 2.71828...$

Which graph do you feel $y = e^x$ will more closely resemble, $y = 3^x$ or $y = 3^{-x}$? Why?

Example 3:

Sketch the graph of $y = e^x$ using a table of values. Specify the domain, range, intercept(s) and asymptote.

x	y



Domain:

Range:

Intercept(s):

Asymptote:

Observe the graph of $y = e^x$. Is the function $f(x) = e^x$ a one-to-one function? Explain. Include the significance of one-to-one functions in your answer.

To sketch the graphs of $y = e^{-x}$, $y = -e^x$, $y = e^x \pm c$ or $y = e^{x \pm c}$, remember your rules concerning reflections and translations.

Summary of Exponential Graphs

Logarithmic Functions

On page two of this set of notes, we decided the function $f(x) = b^x$ is one-to-one and therefore has an inverse function. We will now try to find this inverse algebraically.

Following the four steps outlined in chapter 3, we have:

1)

2)

3) We're 'stuck'. Thus far we have no mathematical operation to solve for a variable exponent.

Rather than 'give up' trying to find the inverse of $y = b^x$, mathematicians of long past decided to interpret the meaning of step 2), above.

'*y* is the exponent on base *b* that yields *x*'

transformed into

'y is the *logarithm* on base *b* that yields *x*'

The key to remember here is that '***logarithm' means 'exponent'***.

Of course, notation soon followed the discovery of 'logarithms'.

For $b > 0 and b \neq 1$

$$x = b^y \Leftrightarrow y = log_b\, x$$

Proficiency in translating from exponential form to logarithmic and vise versa is a must.

Example 4:

Rewrite each given function as its inverse to complete the chart.

Exponential Form	Logarithmic Form
$4^{-2} = \frac{1}{16}$	
	$log_c\, x = y$
	$log_3\, 9 = 2$
$2^{-1} = \frac{1}{2}$	
	$log_x 64 = 2$
$y = b^x$	
	$log_2 \frac{1}{8} = x$
$e^x = y$	
$10^y = x$	

The last two examples bear further conversation:

Keep in mind that a logarithm is an exponent. Pay close attention to the following set of examples.

Example 5:

Solve each of the following for the unknown.

a) $log_x 64 = 2$

b) $log_2 \frac{1}{8} = x$

c) $log 100$

d) $ln 8$

e) $log_8 m = -3$

f) $log_a a^{-2} = x$

g) $log_a a^4 = x$

The last two examples illustrate one of the logarithm properties:

$$log_b b^x = x$$

It follows then, that

$$log 10^x = x \text{ and } ln e^x = x$$

There is a similar exponential property that states $b^{log_b x} = x$

Example 6:

Solve the following:

a) $e^{ln4} = x$

b) $log10^3 = x$

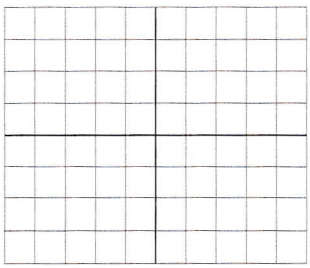
c) $lne^2 = x$

Logarithmic Graphs

Now that you are able to transform an exponential function into a logarithmic function algebraically, let's find the graph of $y = log_b\, x$using the graph of $y = b^x$and our knowledge of the relationship between the graphs of a function and its inverse.

Example 7:

Sketch the graph of $f(x) = b^x$ and its inverse, $f^{-1}(x) = log_b\, x$for b > 1, x > 0 on the same set of axes. Indicate the domain, range, intercept(s) and asymptote for each function.



	$y = b^x$	$y = log_b\, x$
Domain:	_____	_____
Range:	_____	_____
Intercept:	_____	_____
Asymptote:	_____	_____

For the rest of this chapter, we will assume the base, *b*, is greater than one.

Notes for graphs of $log_b(x \pm c)$

- The larger the base of the log, the closer the graph will be to the x-axis.
- To find the domain of a log function, set the argument > 0, and sovle.
- To find the equation of the vertical asymptote, set argument = 0 and solve.
- The range for ALL log graphs is $(-\infty. \infty)$

Example 8:

Give the domain for each of the following. Put your answer in interval notation.

a) $y = log_3(2 + x)$

b) $y = log_2(4 - x)$

c) $y = log_3(2x - 5) + 1$

Example 9:

Give the equation of the vertical asymptote for each of the following.

a) $y = log_3(2 + x)$

b) $y = log_2(4 - x)$

c) $y = log_3(2x - 5) + 1$

We can graph a log function by making an x-y chart. We carefully select values of x so the argument is a power of the base. This will allow use to use the rule $log_b b^x = x$

Example 10:

Make an x-y chart for each of the following

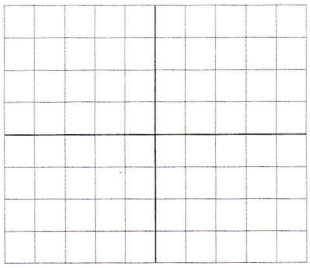
a) $y = log_2(x)$

b) $y = log_4(4 - x)$

On the next page we will do an example that puts all the information together.

Example 10:

Sketch the graph of $y = log_3(4 - x) + 1$. Specify the domain, range, intercept and asymptote.



Domain: _____

Range: _____

Intercept: _____

Asymptote: _____